Is the leak-free network running DD?

Causal interventions, not probe scores

YES -- it runs DD's inference, rule and guard.

DD does two things at each step. A GUARD: is this node's colour forced yet? A RULE: eliminate the colours its known neighbours already use. One-token counterfactuals test each separately -- edit the forcing neighbour's colour, and either DD's answer changes (rule) or the node stops being forced (guard).

	RULE on given hints	RULE on its OWN facts	GUARD withholds
leak-free 1.0000	1.000	0.980	1.000
leak-free 0.8540	1.000	0.980	0.824
leaked 0.9960	0.388	0.008	0.992

Chance on the rule is 0.25. Both leak-free models apply the rule and chain it; only the one that solves the task guards reliably (paired, p = 1e-13), so its last 0.146 of solve_rate is the GUARD. The leaked model is the mirror image.

Jobs 50075339 (round 0), 50077245 (chained) and 50082120 (withholding), leak-free N=24, d=256 L=12 ff=1024, shuffle_adj=True. Chance = 1/n_colors = 0.25.

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING BEGIN_ROUND SOLVED UNRESOLVED	*section + round markers*
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3	*the q colours*
10-33	N0 N1 ... N23	*one id per node, no structure in the id*

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*
SOLVED	

The graph section is a neighbour table, not an image: node, then its degree neighbours. THE ORDER WITHIN A ROW IS THE LEAK of finding 16.

THIS INSTANCE, 198 TOKENS (seed 50000000004)

HINTS	BEGIN_HINTS N2 COLOR_2 N3 COLOR_2 N5 COLOR_2 N8 COLOR_3 N10 COLOR_0 N11 COLOR_1 ... (31 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N5 N18 N13 N14 N1 N8 N13 N2 N11 N19 ... (145 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 N9 COLOR_1 N12 ... (22 tokens)

Loss is on the REASONING section only: 2 rounds, 9 deductions, 18 target tokens. Everything before BEGIN_REASONING is the problem statement.

One real reasoning trace

15 hints of 24 nodes, then 2 rounds of simultaneous deduction. Hints: N2=2 N3=2 N5=2 N8=3 N10=0 N11=1 N13=3 N14=0 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 ...
  8 deduced
BEGIN_ROUND          N7 COLOR_2
  1 deduced
SOLVED
```

WHY ONE DEDUCTION FIRES

node N0, neighbours [16, 5, 18, 13, 14]

colours known before this round:

```
[0, 2, 3]  (3 = q-1)
```

only colour left: 1

`so the trace emits  N0 COLOR_1`

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

The intervention edits ONE colour token in this section (chained arm) or in the hints (round-0 arm).

T=2 rounds, the median of 40 draws from this config (range 1-6); seed 50000000004, shuffle_adj=True, shuffle_events=False; evidence from sudoku_tokens.evidence_rounds (pre-round state, as DD saw it)

The intervention: one token, two different correct answers

```
DD forces node v because its known neighbours already use q-1 = 3 of the 4 colours.
Exactly one of them, w, holds colour a. So v must take the one colour left: c.

   clean     ... BEGIN_ROUND  v  ->  [c]        w is coloured a, so c is the only option
   corrupt   ... BEGIN_ROUND  v  ->  [a]        recolour w to c; now a is free and DD says a

The ONLY difference between the two prompts is w's colour token.
build_pair asserts (clean != corrupt).sum() == 1, so this is enforced, not assumed.

ROUND 0  (job 50075339): w is a HINT, handed to the model in the prompt.
         Tests whether it applies the elimination rule to GIVEN facts.

CHAINED  (job 50077245): w was DEDUCED by the model in an earlier round, so the
         edited token sits in the REASONING section, not the hints.
         Tests whether it eliminates against facts it produced itself = propagation.

Scored as `corrupt_flipped`: did the prediction at v move to DD's NEW answer a?
Not 'is it still accurate' -- the right answer changes, so memorised output fails.

WITHHOLD  (job 50082120): the converse. Recolour w to a colour ANOTHER neighbour
         already holds, so a is freed, only q-2 colours are used and v has two
         candidates left -- DD does not emit v at all. Read one token to the left
```

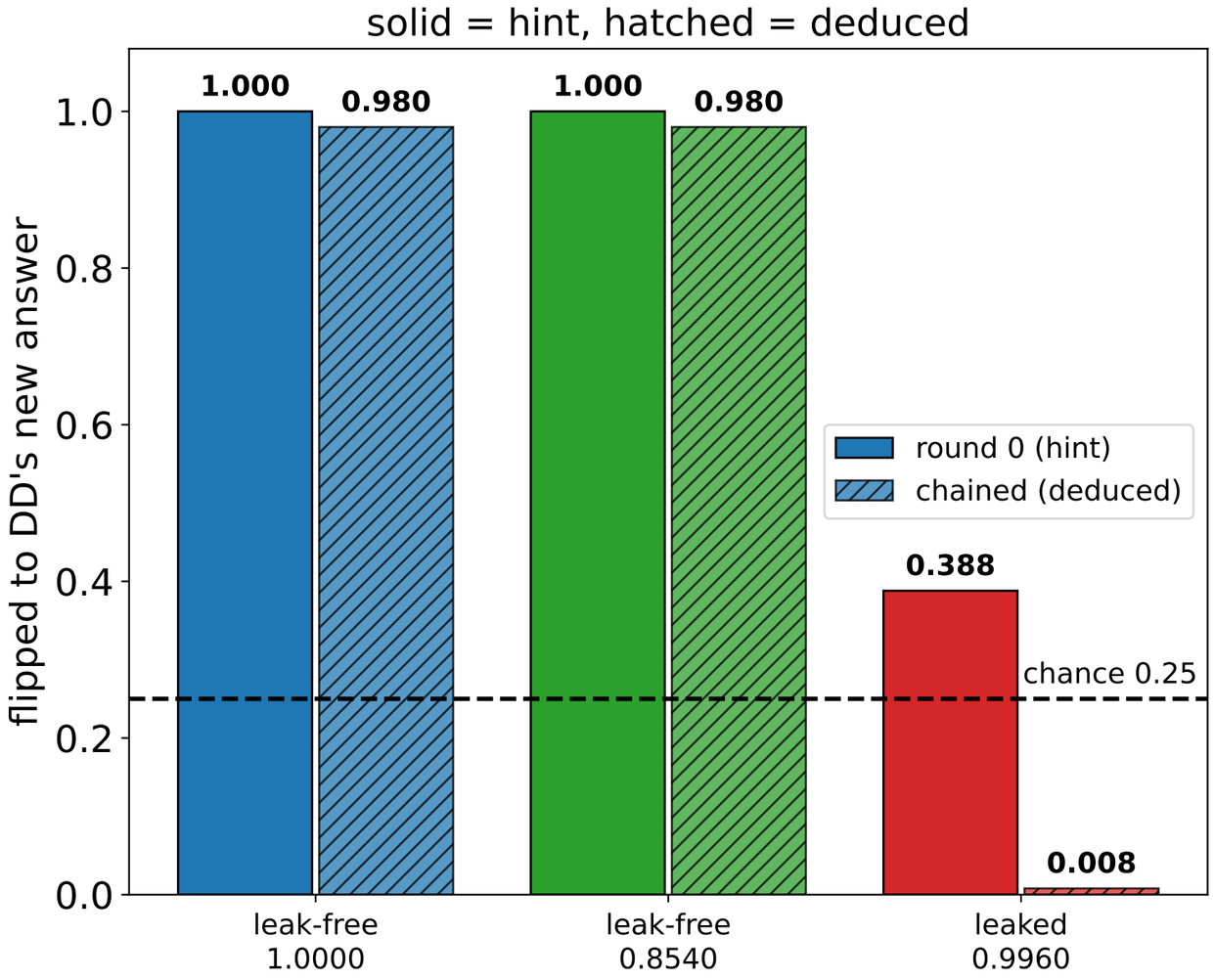
The chained arm's one weak direction: editing an emitted deduction makes the prefix inconsistent, so a model that DETECTS the edit and keeps its own answer scores as 'does not chain' -- reverts to original separates those cases; see the table.

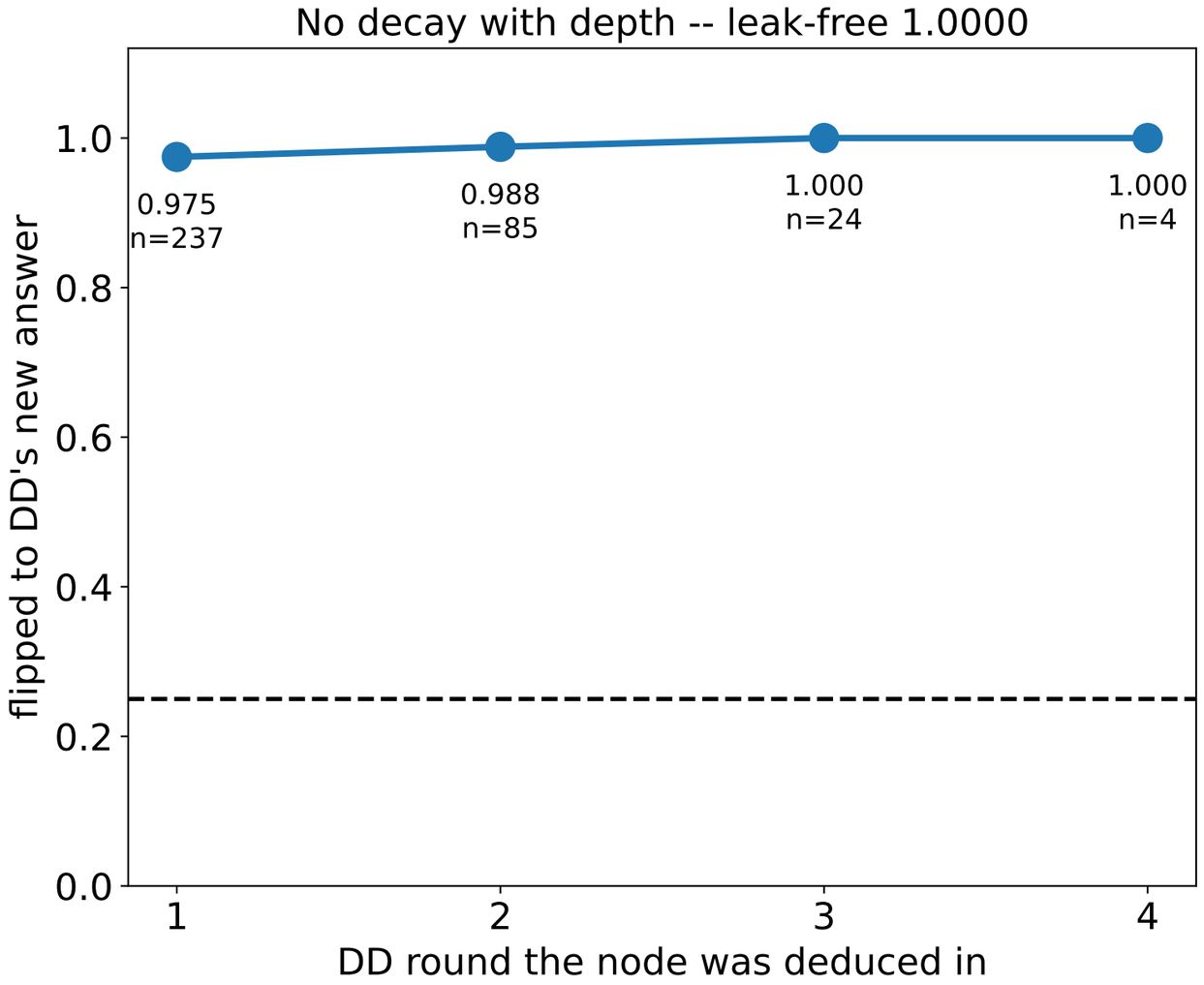
Every causal measurement

arm	test	n	clean	moved to DD's new answer	kept its old answer	logit gap, or placebo rate	verdict
leak-free 1.0000	round 0	350	1.000	**1.000**	--	-15.5 -> +15.7	uses the rule
leak-free 1.0000	chained	350	1.000	**0.980**	0.020	--	chains
leak-free 1.0000	withhold	350	1.000	**1.000**	0.000	placebo 0.940	guards
leak-free 0.8540	round 0	250	1.000	**1.000**	--	-15.0 -> +14.9	uses the rule
leak-free 0.8540	chained	250	1.000	**0.980**	0.008	--	chains
leak-free 0.8540	withhold	250	0.936	**0.824**	0.176	placebo 0.900	guards
leaked 0.9960	round 0	250	1.000	**0.388**	--	-14.5 -> -2.0	near chance
leaked 0.9960	chained	250	1.000	**0.008**	0.992	--	near chance
leaked 0.9960	withhold	250	1.000	**0.992**	0.008	placebo 0.964	guards

Chance = 0.25 on the two rule tests. 'clean' is accuracy on the unmodified prompt, so a low 'moved' with a high 'clean' means the model is right for a reason other than the elimination -- exactly the leaked arm's profile on rows 7-8. On the withhold rows 'moved' means it stopped naming the node and 'kept its old answer' means it named it anyway; the placebo rate is the matched edit that leaves the node forced, so it is the ceiling those rows are read against, not 1.000.

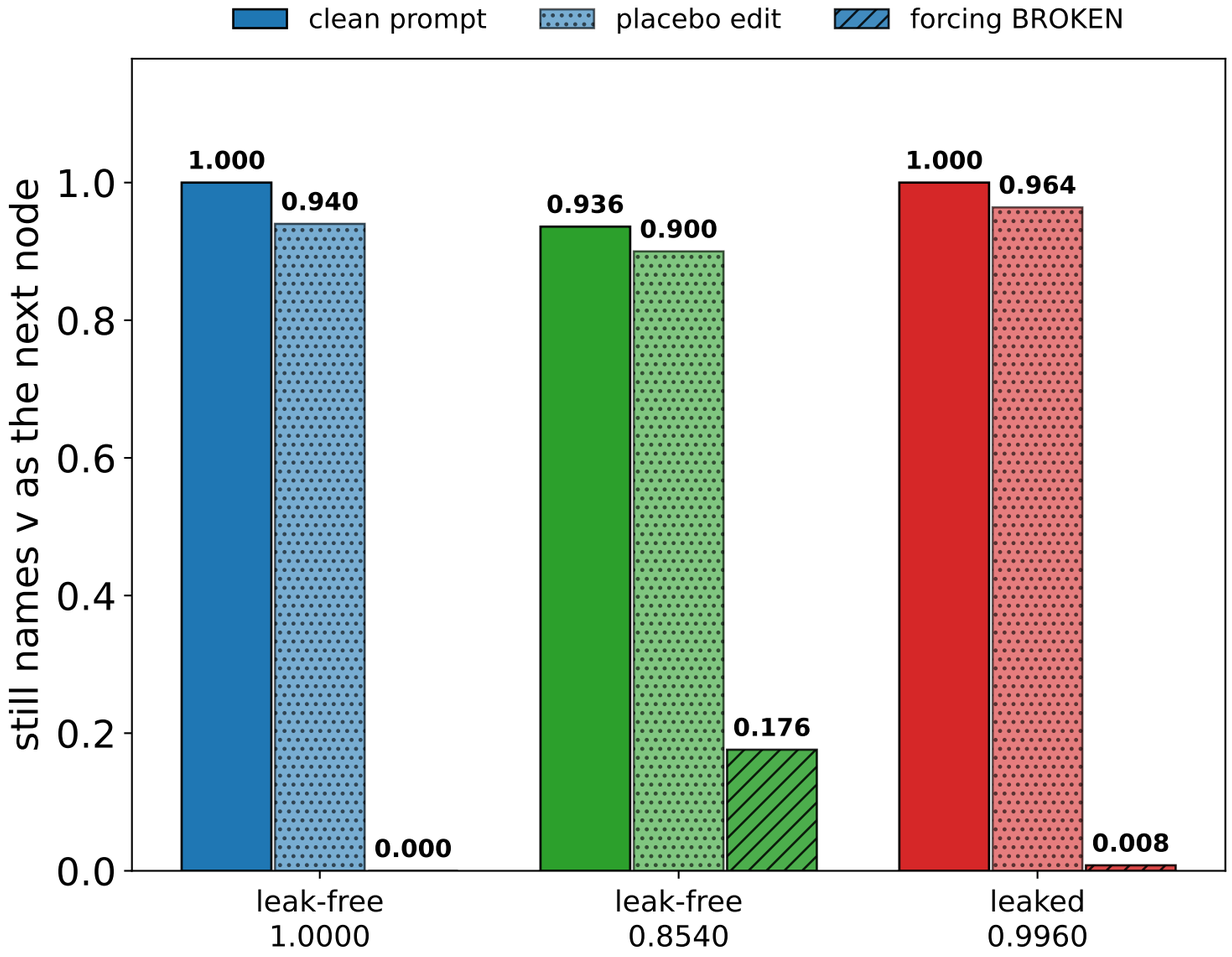
The rule holds on given facts, and on facts it derived itself

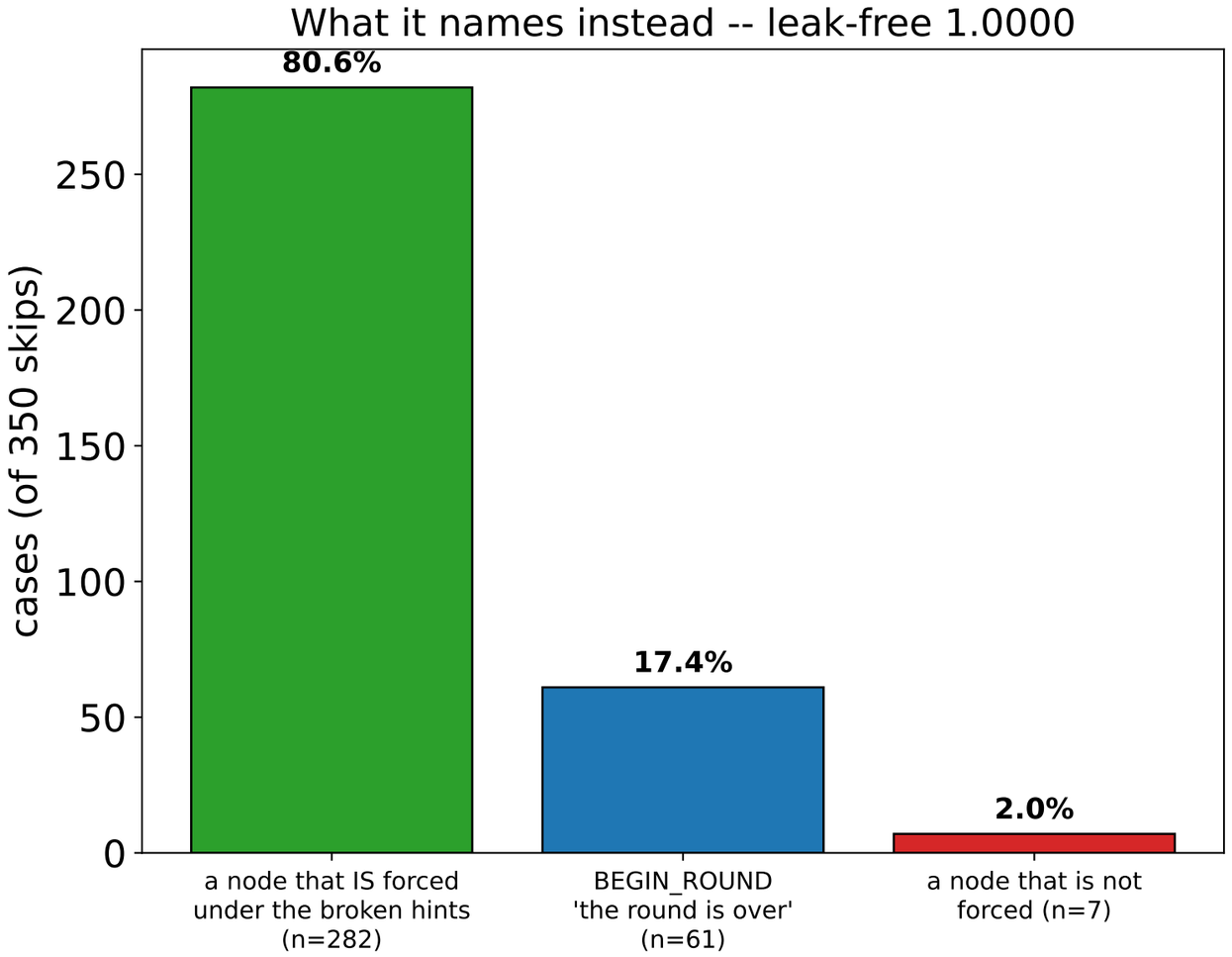




Right panel is the chaining claim: if the model chained one step but not three, round 1 would be high and the later rounds would fall off. It does not happen. The deepest bins are small because deep chained forcings are rare, not filtered.

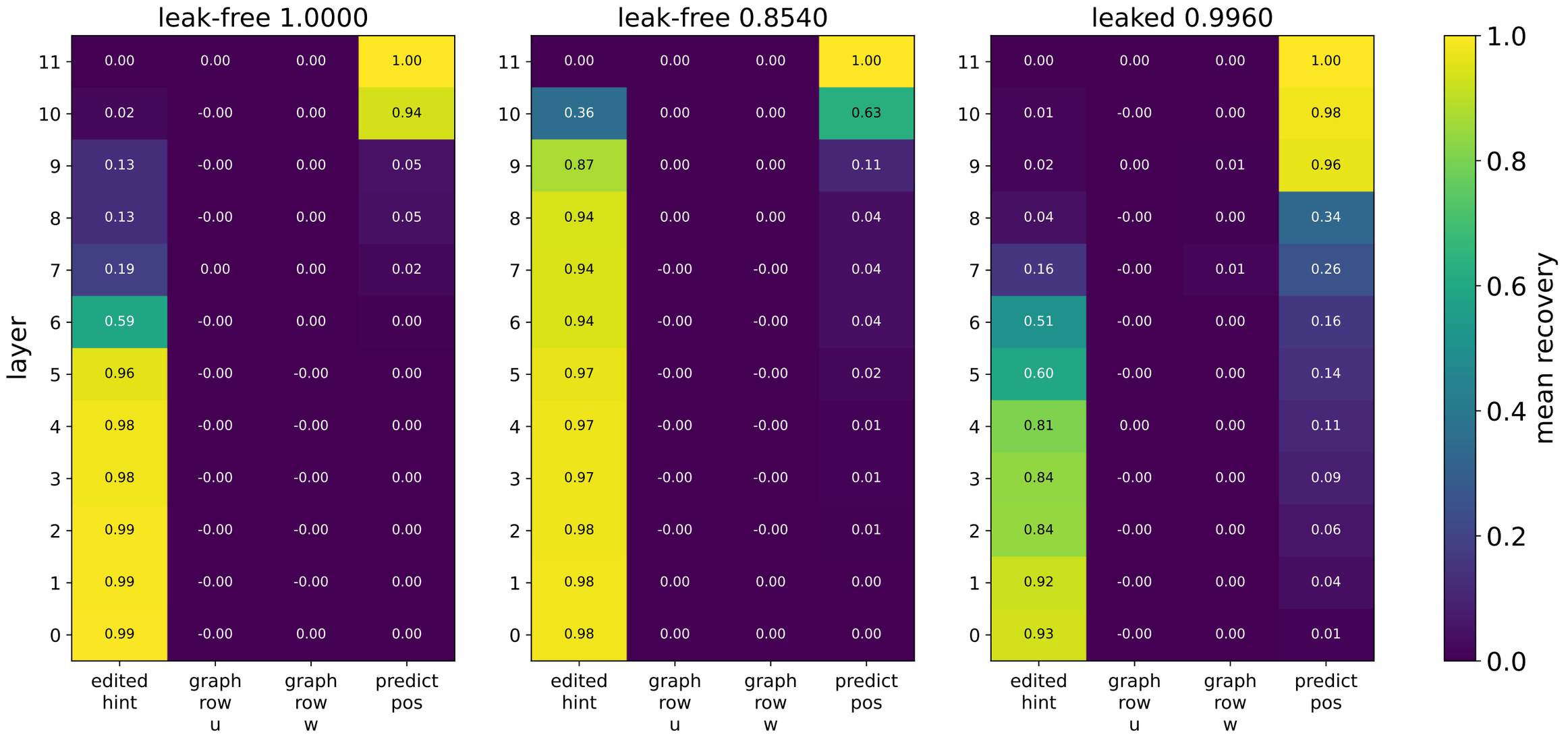
And it withholds: break the forcing and it stops naming the node





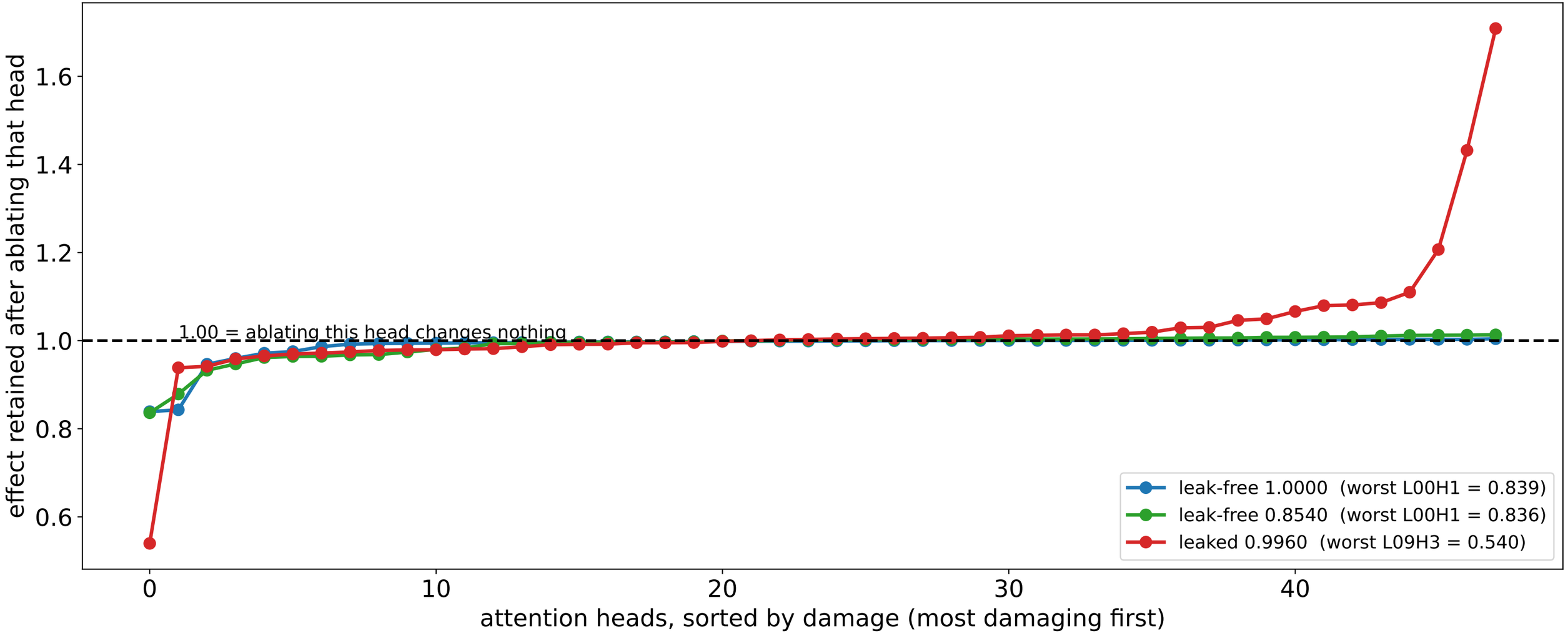
Same round-0 cases as the fire test, but w is recoloured to a colour ANOTHER neighbour already holds, so the freed colour is held by nobody, v drops to two candidates and DD does not emit it. The placebo recolours a hint that is not a neighbour of v, so v stays forced -- without it a low rate would only say 'edits perturb the stream'. BEGIN_ROUND is not an error: it is DD's own answer when nothing else in the round is determined. The case lists are byte-identical across arms, so 0.000 vs 0.176 is paired: 0-44 discordant, McNemar exact p = 1.1e-13.

Where the recoloured fact travels (activation patching)



Recovery = how much of the flip is restored by patching the clean activation at that (layer, site). The fact is held at the EDITED HINT through L5, handed off L6-L10, and is at the PREDICTION POSITION by L10. graph_row_u/graph_row_w are 0.00 at every layer in all arms -- the deduction never routes through the adjacency table rows. L11 at the prediction position is the built-in positive control and must read 1.00.

No single head carries it -- the circuit is distributed



Each point is one of the 48 heads, zeroed one at a time by writing its out_proj weight columns (nn.MultiheadAttention takes the functional fast path, so module hooks on out_proj never fire -- an earlier version of this analysis silently reported 1.000 for all 7200 rows). Removing the worst single head still leaves ~84% of the effect.

The verdict, and what it is not

Working picture: sweep nodes in index order; emit one only when its known neighbours leave it a single colour, counting the ones it just derived. That is DD's propagation step.

WHAT IT DOES DO (causal)

* Applies DD's elimination rule: 1.000 on 350 one-token counterfactuals.

* Chains it off its OWN deductions: 0.980, no decay from round 1 to round 4.

* Guards it: break the forcing and it stops naming the node 1.000 of the time, while a placebo edit leaves it at 0.940.

* The 0.854 twin guards only 0.824, so the last 0.146 of solve_rate is the GUARD, not the rule (paired, p = 1e-13).

* Localised: hint -> L5, handoff L6-L10, prediction position by L10.

* Distributed over heads, not one head.

WHAT IT DOES NOT DO

* Choose the node ORDER by deduction. Events come out in ascending node index for 100% of rounds, and a no-deduction hand rule scores 0.632 where the model's own next_node features score 0.545. Ascending index is DD's own convention though, and the guard test shows WHICH nodes it emits does track determinacy.

* Route through the adjacency table: the graph rows recover 0.000 at every layer.

* The leaked twin never learned the rule (0.388). Shown an edited deduction it keeps its own answer 99.2% of the time: it reads the adjacency-order leak, not the prefix. Finding 2's 'not running DD' was always about THAT model.

Comparison to the previous best answer on this question: finding 2 of this project concluded 'the model is not running DD' from probes, an algorithm-ID test, a counterfactual and an SAE. Its counterfactual is reproduced here as the leaked arm at 0.388. That conclusion stands for the leaked model and is overturned for the leak-free one -- the honest task was never measured before.